THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 6: **Perimeter and Area** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The area of a quadrilateral whose diagonal is 24 cm and whose offsets from the diagonal are 9 cm and 11 cm is (cm^2): 1

(a) 240 **(b)** 241 **(c)** 239 **(d)** 242

2. A rectangle of perimeter 72 cm has length 26 cm. Its area (cm^2) is: 1

(a) 262 **(b)** 260 **(c)** 259 **(d)** 261

3. A circular wire of radius 35 cm (take $\pi = \frac{22}{7}$) is bent into a square. The side of the square (cm) is: 1

(a) 56 **(b)** 57 **(c)** 54 **(d)** 55

4. A wheel of diameter 42 cm (take $\pi = \frac{22}{7}$) makes 100 revolutions. The distance covered (in metres) is: 1

(a) 131 **(b)** 132 **(c)** 133 **(d)** 134

5. The length of the arc of a sector of angle 80° in a circle of radius 63 cm (take $\pi = \frac{22}{7}$) is (cm): 1

(a) 89 **(b)** 87 **(c)** 90 **(d)** 88

6. A triangle has area 36 cm^2 and one side 9 cm. The altitude on this side (cm) is: 1

(a) 8 **(b)** 7 **(c)** 9 **(d)** 10

7. The height of a parallelogram of area 234 cm^2 and base 26 cm is (cm): 1

(a) 10 **(b)** 8 **(c)** 9 **(d)** 11

8. A square has area 1024 cm^2. Its perimeter (cm) is: 1

(a) 127 **(b)** 128 **(c)** 130 **(d)** 129

9. The area of a quadrilateral whose diagonal is 30 cm and whose offsets from the diagonal are 10 cm and 4 cm is (cm^2): 1

(a) 212 **(b)** 210 **(c)** 209 **(d)** 211

10. A rectangle of perimeter 84 cm has length 30 cm. Its area (cm^2) is: 1

(a) 362 **(b)** 361 **(c)** 360 **(d)** 359

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the arc length for 240° in a circle of radius 21 cm (take $\pi = \frac{22}{7}$). 2

12. A triangle has sides 11 cm, 25 cm and 30 cm. Find its area and the altitude on the side 30 cm. 2

13. Find the area of a quadrilateral $ABCD$ in which diagonal $AC = 28$ cm and the perpendiculars from B and D to AC are 5 cm and 9 cm. **2**

14. Find the arc length for 150° in a circle of radius 42 cm (take $\pi = \frac{22}{7}$). **2**

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the area of the cyclic quadrilateral with sides 4 cm, 14 cm, 18 cm and 28 cm. **3**

16. A rectangular sheet 21 cm $\times$ 14 cm has a semicircle of diameter 14 cm added on one breadth (take $\pi = \frac{22}{7}$). Find the perimeter and the area of the whole figure. **3**

17. The sides of a parallelogram are 30 cm and 25 cm and the height on the side 30 cm is 10 cm. Find its area and the height on the side 25 cm. **3**

OR

Find the area of the cyclic quadrilateral with sides 6 cm, 8 cm, 25 cm and 27 cm. **3**

18. A rectangular sheet 50 cm $\times$ 28 cm has a semicircle of diameter 28 cm added on one breadth (take $\pi = \frac{22}{7}$). Find the perimeter and the area of the whole figure. **3**

OR

The sides of a parallelogram are 14 cm and 21 cm and the height on the side 14 cm is 12 cm. Find its area and the height on the side 21 cm. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A rectangular plot 80 m $\times$ 50 m has a circular pond of radius 21 m in it (take $\pi = \frac{22}{7}$). Find the area of the plot left, and the cost of turfing it at ₹3 per m^2. **5**

OR

A triangular field has sides 12 m, 17 m and 25 m. Find its area, and the cost of ploughing it at ₹6 per m^2. **5**

20. A rectangular plot 70 m $\times$ 42 m has a circular pond of radius 14 m in it (take $\pi = \frac{22}{7}$). Find the area of the plot left, and the cost of turfing it at ₹7 per m^2. **5**

OR

A triangular field has sides 20 m, 21 m and 29 m. Find its area, and the cost of ploughing it at ₹2 per m^2. **5**

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND